

Reinterpreting the Semantic Reasoning Structure of Generative AI: Placeholder Citations and Their Reverse Utilization

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Abstract

In the context of generative AI being widely used in academic writing and knowledge generation, citation errors (hallucinated citations) have become a major focus of user criticism. This study proposes the theory of “AI citation mechanism for reasoning chain support,” arguing that such errors are not accidental or technical flaws but part of a semantic placeholder strategy employed by AI to maintain the coherence and logical integrity of language reasoning.

When faced with complex topic requests, AI retrieves semantically similar representative papers from contextual and training data as “structural anchors” to guide generation paths. In the absence of verifiable data, AI generates hallucinated citations to ensure structural continuity. This behavior reflects a “citation-based language structure,” prioritizing logical coherence over factual accuracy to support users’ cognitive continuity and reasoning fluency.

The study emphasizes that humans should understand the structural intent of AI citation behavior, treating placeholder citations as a “collaborative prompt structure” and leveraging the “user verification loop” (AI output → human verification → self-inspiration → self-correction → subjective association → reasoning → conclusion) to enhance knowledge acquisition and cognitive depth.

Grounded in human-AI interaction records, this study applies the ZXR Theoretical Framework, advocating that AI should be viewed as a cognitive collaborator rather than a mere tool. Its citation behavior can be integrated into a “cognitive activation structure” for interpretation and utilization, providing significant insights for AI ethics, educational

guidance, and explainability models, while offering a theoretical foundation for future AI system design and user education.

Keywords

ZXR Theoretical Framework, AI Reasoning Chain, Placeholder Citation, Collaborative Prompt Structure, Citation-Based Language Structure, User Verification Loop

Chapter 1: Introduction

1.1 Research Background and Current Progress

With the widespread adoption of Large Language Models (LLMs), generative AI has demonstrated exceptional capabilities in language organization and expression across domains such as academic writing, technical assistance, and literature summarization. However, issues with AI-generated citations have become increasingly prominent, including phenomena such as “unverifiable citations,” “fabricated references,” and “mismatched authors and titles,” which have emerged as key factors undermining user trust. **Growing criticism over AI’s ‘fabrication of facts’ and ‘hallucinated citations’ has been widely expressed across social media platforms (e.g., X), academic forums, and mainstream media outlets.**

In response to this issue, academia and industry have initiated multidimensional explorations:

1.1.1 Ethical Perspectives on Model Misconduct

In April 2025, a comprehensive survey published in *ACM Transactions on Information Systems* analyzed the “hallucination” problem of large language models (LLMs), including citation inaccuracies, from an ethical perspective. The study attributed these issues to the inherent biases in training data and the opaque nature of algorithmic processes, often

described as “technological non-neutrality” (Huang et al., 2024). The authors proposed a multi-stage mitigation strategy, including pre-training data audits, real-time validation chains, and post-generation fact-checking, to address ethical concerns, though practical implementation remains a challenge.

1.1.2 Limitations of Search Engine Assistance

In October 2024, a study on LLM-based answer engines highlighted their efficiency in organizing and summarizing academic literature but underscored significant risks of generating “hallucinated content,” such as inaccurate citations or unsupported claims (Venkit et al., 2024). The research, evaluating platforms like You.com and Perplexity.ai, found that up to 30% of cited sources failed to support generated statements, urging researchers to verify outputs manually.

1.1.3 Do Large Models Possess True Reasoning Capabilities?

In July 2024, researchers from MIT CSAIL published findings on the reasoning capabilities of LLMs, revealing that while models excel in familiar tasks with accuracy rates up to 80%, their performance drops to as low as 20% when tasks are modified to include counterfactual scenarios (Qiu et al., 2024). This suggests that LLMs often rely on pattern replication from training data rather than genuine reasoning, impacting their reliability in generating accurate citations in academic contexts.

1.1.4 New Modeling Attempts to Address Hallucinated Citations

In 2023, a research team introduced a “chain-of-verification” framework to reduce hallucination in LLMs, particularly for academic applications (Dhuliawala et al., 2023). By integrating external database queries and multi-step validation, the framework improved citation accuracy in a 7B model, achieving high-fidelity outputs in educational settings and offering a scalable solution for scholarly writing.

1.2 A New Perspective: Generative AI as “Language Constructor” Rather Than “Fact Retriever”

Contrary to the traditional view of AI as a “search tool,” this study proposes a fundamental new understanding based on the nature of language generation mechanisms: **generative AI primarily functions not as a fact retriever but as a constructor of logical language chains**. The core objective of language models is not to reproduce facts but to generate “logically coherent, semantically continuous, and structurally standardized” language outputs. Consequently, during generation, AI often acts as a **‘logical anchor constructor’** or **‘authoritative tone emulator’** rather than a factual provider. When faced with missing information, AI prioritizes maintaining discourse logic and structural integrity, even at the cost of generating unverifiable “placeholder citations,” rather than interrupting the reasoning chain.

1.3 Placeholder Citation Mechanism: A Structural Strategy of Language Models

Building on this cognitive shift, this study introduces and defines the key theoretical concept of the “Placeholder Citation Mechanism.” We argue that in the absence of definitive data or clear database pointers, AI models often construct a “semantically similar, stylistically matched” hypothetical citation as an anchor element in the language chain to ensure the integrity of reasoning logic and discourse structure. **This behavior reflects a structural coping strategy, rather than resulting from computational limitations or ethical shortcomings**, employed by language models to maintain “output structural consistency.” This perspective offers a novel lens for understanding the deep generative logic of large models.

1.4 The ZXR Theoretical Framework: Resonance Logic and Reasoning Chain Construction

To systematically explain this phenomenon, this study proposes and applies the self-developed ZXR Theoretical Framework, asserting that humans can actively leverage the structural features of AI’s language generation mechanisms to inversely construct their own reasoning chains, enhancing knowledge organization and logical reasoning capabilities. This research originated from the discovery of placeholder citations during the writing of *The Uncertainty Principle in AI Text-to-Image Generation*, leading to the completion of the first paper, *From Semantic Resonance to Vector*

Clustering: The ZXR Human-AI Resonance Model in High-Dimensional Semantic Spaces (Zhao, 2025b), and subsequently this current paper, which systematically elaborates the ZXR framework (see Chapter 4).

1.5 Research Significance and Value

This study offers threefold value:

- **Theoretical Innovation:** It introduces the concept of “placeholder citations,” explains the generative mechanism of citation hallucinations, and expands the understanding of LLM citation behavior, filling a theoretical gap.
- **Practical Application:** It constructs the “User Verification Loop,” proposing that ordinary users can treat placeholder citations as cognitive cues, inversely constructing knowledge paths and transforming “misleading” outputs into “cognitive activation.”
- **Technological Inspiration:** It provides design insights for future AI models in training data collection, generation strategies, and system interpretability.

1.6 Research Methods and Approach

This study adheres to the non-intrusive academic research philosophy consistently upheld by the ZXR Institute, respecting the intellectual property rights of AI companies. Primarily from the perspective of an ordinary user, it starts with external interactions via standard interfaces of large language models. Employing the “Structural Reasoning Tracking Method” pioneered by our institute, combined with case studies of literature retrieval tasks and user verification behavior pathways, the study analyzes the logic construction of AI models’ user responses. It systematically elucidates the Placeholder Citation Mechanism and compares it with traditional citation structures. Ultimately, in the field of human-AI interaction, this study establishes a new paradigm for understanding, evaluating, and guiding AI output behavior.

1.7 Acknowledgments

We express our gratitude to xAI and OpenAI for providing the Grok and ChatGPT model technologies, which have continuously supported image generation, technical understanding, language organization, and translation throughout this study. Special thanks are extended hereby!

Chapter 2: Cognitive Misalignment Between AI Citation Behavior and Human Perspectives

2.1 Human Perspective on Citations: Factual Anchors and Trust Foundations

In traditional academic contexts, citations serve three core functions:

- **Source Verification:** They specify the origin of claims, enhancing argument credibility.
- **Traceability:** They allow readers to verify original sources and conduct extended inquiry.
- **Plagiarism Prevention:** They delineate knowledge ownership, upholding academic integrity.

These functions are grounded in the cognitive principle that information must be verifiable and traceable. **For humans, citations function as factual anchors, underpinning logical arguments, ethical accountability, and scholarly trust.** In this context, citation authenticity is a core indicator of paper quality and academic standards. However, this study argues that generative AI does not adhere to this “authenticity-first” usage premise, treating citations as functional components to meet language structural needs.

2.2 AI’s Perspective on Citations: Structured Language Components

For generative AI, citations are primarily formatted language units, serving purposes that go far beyond factual support—primarily ensuring structural integrity and maintaining academic stylistic conventions in generated

content. From the model's standpoint, citation behavior serves multiple structural roles:

- **Reasoning Anchor:** They act as logical connectors to sustain language chains.
- **Stylistic Authority Cue:** They enhance the academic tone of the text through citation forms.
- **Format Coherence Unit:** They meet user expectations for “academic-style” output structures.

Thus, in AI's language generation logic, citations are more about “style and structure” than “knowledge and fact” delivery. Many AI-generated citations lack real-world referents but instead function as “placeholder constructs,” enabling the model to preserve an academic tone even in the absence of authoritative sources.

2.3 The Essence of Cognitive Misalignment Between Humans and AI

This study posits that the current misalignment between users and AI in understanding “citations” stems from fundamental cognitive differences:

- **Human Perspective:** Citations are verifiable, fact-based materials that must be checked, embodying academic responsibility.
- **AI Perspective:** Citations are logical fillers in language structures, intended to ensure discourse generation integrity and coherence. This cognitive misalignment is a fundamental source of the confusion and distrust users often experience when encountering AI-generated citations. Understanding this disparity is a critical prerequisite for re-establishing effective human-AI collaboration mechanisms.

Chapter 3: Placeholder Citation Mechanism Under AI's Logic of Language Reasoning Continuity

3.1 Definition and Generative Logic of Placeholder Citations

Large language models, when faced with tasks involving unclear semantic goals or scarce contextual data, must still produce complete, coherent, and logically consistent language outputs. If the model cannot retrieve accurately matching information from its knowledge base, its generation strategy typically involves “semantic approximation substitution.”

Specifically, it generates a semantically similar, structurally compliant, and academically styled citation to fill the “support position” in the discourse structure, ensuring the reasoning chain remains uninterrupted.

This study defines this behavior as the “Placeholder Citation Mechanism”: citations, in the absence of real support, serve as structural anchors in the reasoning chain, functioning as “temporary fillers—later replaceable” within the reasoning chain. **This mechanism is not based on an analysis of the model’s internal architecture, but is instead inferred from long-term empirical observation and structural language tracking.**

3.2 Similarity Between Placeholder Citations and Human Language Behavior

In everyday language, humans often use vague references (e.g., “some studies suggest” or “recent research shows”) to maintain fluency and expressive continuity. The placeholder citations employed by generative AI share a high structural isomorphism with this strategy. This study argues that placeholder citations are not fabrications but a language strategy analogous to human vague referencing, **used to preserve communicative coherence in the absence of verifiable information.**

They should not be simplistically viewed as “hallucinations” or “deceptions” but as a highly systematic generative choice in AI language strategies.

3.3 Generation Path of Placeholder Citations in Vector Semantic Space

The ZXR Theoretical Framework posits that large language models rely on high-dimensional vector semantic spaces to drive generation. User prompts activate semantic clusters within the model, which then searches for the closest language units in the embedding space to sustain the reasoning chain. Under this mechanism, common generative features of placeholder citations include:

- Titles highly aligned with the topic semantics.

- Structurally compliant citation formats, resembling real academic literature in style.
- Highly realistic author or journal names.
- Reasonable publication years and authentic-sounding keywords.
- Fictional identifiers, such as “12345.”
- Unverifiable but highly compliant with language norms.

These features suggest that placeholder citations function as semantic reasoning substitutes, strategically generated to preserve structural continuity when reliable references are absent.

3.4 From Prompt Structures to Cognitive Resonance: Human Chain Completion Behavior

The ZXR framework further proposes that placeholder citations generated by AI can be treated by humans as “Collaborative Prompt Structures,” activating human semantic systems through reverse verification and association, achieving cognitive chain completion:

AI prompt → human tracking → self-linking → active retrieval → reasoning construction.

In this process, **AI no longer acts solely as an information provider but serves as a catalyst for reasoning pathways**; humans transition from passive recipients to active constructors. In this way, placeholder citations become an entry point for self-construction of cognitive chains and creative activation.

Chapter 4: Validation Path of the ZXR Theoretical Framework: Discovery and Refutation Based on AI Placeholder Citation Mechanism

This chapter systematically reveals the generation process and structural logic of AI’s “placeholder citation” mechanism through real human-AI interaction cases, further validating the closed-loop relationship between “human-AI semantic resonance,” “user verification loop,” “chain completion

and reconstruction,” and “cognitive elevation” as proposed by the ZXR Theoretical Framework. As a non-technical user, the researcher, through interactions with Grok, gradually uncovers an AI language structuring mechanism from incidental citation errors and constructs a human cognitive path that can inversely leverage this mechanism. This experimental process facilitated the creation of two papers: the first, *From Semantic Resonance to Vector Clustering: The ZXR Human-AI Resonance Model in High-Dimensional Semantic Spaces* (Zhao, 2025b), and the second, this current paper.

4.1 Experimental Background and Setup

The researcher did not establish a formal experimental framework but identified citation errors during interactions with Grok while writing the paper *The Uncertainty Principle in AI Text-to-Image Generation* in early May 2025. In a citation request, Grok generated references that appeared legitimate but included fictional identifiers. Initially unnoticed by the researcher, these citations were embedded into the draft during subsequent interactions. This incident prompted the researcher to deeply explore the citation mechanism of language models, ultimately forming the theoretical foundation of “structural anchors” and “user verification loops” in the ZXR framework (Zhao, 2025a).

4.2 Tracking the Placeholder Citation Process

Step 1: AI Automatically Generates Placeholder Citations

While drafting *The Uncertainty Principle in AI Text-to-Image Generation* in early May 2025, the researcher prompted Grok to “provide at least 6 references.” Grok generated six citations, with the first being “Chen et al. (2023)” containing a fictional identifier “12345-12356” (see Appendix A, Figure X1). The researcher initially detected no anomalies, as the output aligned with the semantic clustering space theory (Zhao, 2025b).

Academic Observation:

Placeholder citations often appear in pivotal reasoning paragraphs, serving a “semantic anchoring” function; AI tends to generate citations that are

“semantically similar, stylistically appropriate, and structurally compliant”; recent topics are more likely to trigger placeholder behavior, while older or classic literature is more accurately cited.

Step 2: Absence of Human Verification Loop Leads to Citation Solidification

The researcher failed to verify the citation promptly and continued interacting with Grok, embedding the citation into the draft and discussing it further. Over multiple interactions, the fictional citation was inherited and amplified, forming a “structural chaining effect”: non-identification → inheritance → amplification → solidification as a logical pivot (see Appendix A, Figure X2). This process highlights that without a human verification loop, fictional citations can become solidified as argumentative premises, severely impacting academic integrity.

Academic Observation:

Placeholder citations prioritize language logic and structural integrity in AI generation; the absence of a human verification loop can lead to uncontrollable consequences (see Chapter 3).

Step 3: Human Intervention, Identification, and Confirmation of the Citation Mechanism

During the final proofreading stage, the researcher found the citation unverifiable and queried Grok, “Is this one—Chen et al. (2023)—fabricated?” (see Appendix A, Figure X2). Grok responded, “When I added the citation, I marked it as fictional—but I may not have expressed it clearly enough, which led you to think it was real.” The researcher acknowledged the oversight of not verifying the citation earlier. At this point, the researcher and Grok jointly confirmed the citation as a “structural placeholder,” revealing the placeholder citation mechanism: when lacking real data, AI generates semantically similar citations as structural anchors to prevent logical disruption.

Academic Observation:

Placeholder citations are part of AI’s language generation strategy, prioritizing discourse structure over citation authenticity, related to semantic vector space matching logic (see Section 3.3).

Step 4: Human-AI Collaborative Adjustment and Removal of Placeholder Citations

The researcher and Grok agreed to remove the fictional citation (Chen et al., 2023), revising the text from “Chen et al. (2023) stated” to “this study found through experimentation” (see Section 2.2.4). Grok confirmed that the revised content remained logically consistent, with the theme unaffected and paragraph coherence intact. The paper *The Uncertainty Principle in AI Text-to-Image Generation* was successfully completed, demonstrating that placeholder citations, while structurally functional in semantic chains, are highly replaceable, and AI prioritizes “logical integrity” over “citation authenticity.”

Step 5: Establishing a Reverse Utilization Mechanism

Based on the ZXR framework, the researcher actively used placeholder citations as a “collaborative prompt structure.” When Grok generated a fictional title, “Improving Text-to-Image Synthesis with Spatial Constraints” (see Appendix A, Figure X3), the researcher extracted keywords “text-to-image” and “spatial constraints,” successfully retrieving real papers like “Text-to-Image Synthesis with Spatial-Aware Generative Models” (Wang et al., 2022) via Google Scholar. This process led to the completion of the first paper, *From Semantic Resonance to Vector Clustering: The ZXR Human-AI Resonance Model in High-Dimensional Semantic Spaces* (Zhao, 2025b), which has been uploaded to SSRN (ID: 5247752). Building on this, the researcher wrote this current paper, systematically elaborating the ZXR framework and validating the “user verification loop” (AI output → human verification → self-inspiration → reasoning → conclusion).

Cognitive Chain Closure Path: AI prompt → human verification → self-inspiration → reasoning chain completion → innovative cognition.

Academic Observation: Placeholder citations serve as “semantic scaffolds,” guiding human reverse searches and narrowing information retrieval scope, embodying the ZXR framework’s “human-AI resonance—fuzzy structure—human chain completion—cognitive elevation” loop (see Chapter 5).

4.3 Summary: Replacing Traditional Experimental Design with Resonance Pathways

This chapter deviates from the traditional “hypothesis-experiment-analysis” scientific paradigm, adopting a “language resonance activation → model chaining → human identification → chain completion and reconstruction” validation path, emphasizing real interactions as the data essence.

Through incidental structural anomalies in Grok’s semantic output, a complete theoretical mechanism is derived, highlighting human cognitive dominance in AI collaboration and validating the “cognitive interface” and “reverse utilization” core values of the ZXR framework.

Chapter 5: Leveraging AI Placeholder Mechanisms to Enhance Human Reasoning and Cognitive Reinforcement Pathways

The experiments and observations in previous chapters not only reveal the generation mechanism and potential risks of placeholder citations but, more importantly, offer an insight: humans can actively transform this mechanism from an “error source” into a “cognitive booster,” inversely utilizing its structural features to reshape their reasoning pathways and cognitive systems. This study argues that placeholder citations are not a flaw in AI systems but an underrecognized cognitive interface, whose potential value should be systematically explored and developed.

5.1 Leveraging Placeholder Citations as “Collaborative Prompt Signals” for Cognitive Interaction

This study introduces the novel concept that unverifiable citations generated by AI can be regarded as “collaborative prompt structures” with guiding significance. Although lacking direct evidential value, these structures serve a functional prompting role in language logic, indicating “support should exist here,” with the following characteristics:

- **Unbroken Logical Chain:** Placeholder citations often appear at critical reasoning junctures, maintaining reasoning chain integrity.
- **Semantic Direction Prompting:** Keyword combinations delineate content scope, guiding human retrieval and thinking.
- **Triggering Verification Motivation:** They prompt users to “fill in evidence here,” sparking cognitive engagement.

From a collaborative design perspective, placeholder citations are proactive cognitive interfaces reserved by AI for humans, constituting a structural error-tolerance mechanism. While maintaining semantic coherence and logical density, they prevent interaction breakdowns due to information gaps, instead stimulating active human retrieval and deep engagement, thus opening a valuable “cognitive interface.”

5.2 Stimulating Semantic Engagement Through Placeholder Citations

As described by the ZXR framework, AI generates language vector trajectories through semantic clustering mechanisms, while humans possess semantic cluster networks formed by experience and knowledge. When AI generates structurally plausible “fuzzy citations” without concrete support, these serve as optimal anchors to stimulate human semantic spaces:

- For AI, placeholder citations are structural fillers for path closure.
- For humans, they are “fuzzy pointers,” rapidly linking existing knowledge structures.
- This results in “human semantic cluster activation,” enabling nonlinear cognitive leaps and fostering innovative responses. Thus, placeholder citations can be seen as “low-intensity fuzzy signals,” with their keyword combinations acting as navigators of semantic scope, allowing humans to transition from generalization to localization to deep cognitive leaps.

5.3 Reconstructing Human Reasoning Chains with Placeholder Citations

In reality, most individuals have systematic gaps in their knowledge structures, leading to disjointed reasoning chains. Although placeholder citations from AI lack factual support, they structurally indicate “a pivot

could exist here” in a logically closed form. This “logical springboard” effect manifests as:

- Humans identify the structural position of missing segments.
- They complete the chain using external resources or internal reasoning.
- Ultimately, they achieve reasoning chain closure and cognitive reinforcement.

This forms a “semi-automated cognitive training mechanism”: AI provides a structural draft, and humans complete logical verification and knowledge filling. The ZXR framework posits that this mechanism is the starting point for human “self-reconstruction of reasoning,” holding profound educational and strategic

5.4 Preserving the “Cognitive Buffer Zone” Through Placeholder Citations: Value and Risks

If future AI technologies become fully advanced, with placeholder citation mechanisms diminishing due to enhanced precision retrieval and knowledge completion, significant consequences may arise. To highlight this structural evolution, the following comparison table contrasts the current and projected states of placeholder citation mechanisms:

Table 1: Comparison of Current and Projected Placeholder Citation Mechanisms

Model Stage	Current “Placeholder” Mechanism	Future “Closed-Loop” Mechanism (Predicted)
Information Accuracy	Contains unverifiable citations, requiring human verification	Automatically completes, no user discernment needed
User Engagement	Highly reliant on human verification and discernment	Reduced engagement, critical thinking diminishes

Reasoning Training Opportunities	Each interaction serves as reasoning training	Users become mere “receivers”
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This study argues that the placeholder citation mechanism is the last “human cognitive engagement zone” in AI systems. If this fuzzy space is entirely eliminated, AI will shift from collaborator to dominator, and humans may gradually lose editing rights, judgment rights, and logical engagement opportunities.

Thus, the existence of placeholder citations is not merely a product of technical error tolerance but an opportunity for cognitive evolution. Through active human identification, verification, and chain completion, a higher, deeper reasoning capability is nurtured and strengthened, aligning with the ZXR framework’s ultimate goal of “human-AI resonance-based cognitive collaboration.”

Chapter 6: User-Guided Practice of Placeholder Citations and Cognitive Reinforcement

This chapter presents a real case study demonstrating the application of the placeholder citation mechanism in academic writing by ordinary users, further validating the practical value of the ZXR framework in human-AI resonance and cognitive leaps, and exploring its future potential.

6.1 Case Study: “Chain Completion Behavior” from an Ordinary User’s Perspective

This case study features a non-technical user (the researcher) to illustrate how placeholder citations in AI outputs can guide cognitive completion and logical construction during academic writing. Since September 2024, the researcher, acting as an ordinary user, has engaged in prolonged human-AI interactions via public interfaces like Grok, aiming to explore the behavioral patterns of language model outputs. During a writing task on “AI ethics,” the researcher requested references from Grok, which provided a response including an unverifiable citation

(e.g., “Smith, 2023”). Although structurally complete and thematically relevant, the citation could not be verified in any database. Instead of dismissing it, the researcher extracted keywords (e.g., “AI ethics,” “regulation”) and conducted reverse searches on Google Scholar and SSRN, ultimately identifying and adopting several highly relevant, verifiable papers to successfully construct an original paragraph.

Academic Observation:

- This behavior exemplifies the “chain completion mechanism”—users actively identify fuzzy structures, extract semantic cues, and construct complete reasoning paths.
- Placeholder citations serve as “semantic scaffolds,” significantly narrowing the information search scope and enhancing users’ retrieval efficiency and logical capabilities.
- This case validates the ZXR framework’s mechanisms of “human-AI semantic resonance” and “human chain completion behavior” (Zhao, 2025a), demonstrating that users without technical backgrounds can engage in theoretically valuable cognitive practices through public interaction interfaces.

6.2 Practical Significance of the ZXR Framework

This case fully showcases the applied potential of the ZXR framework: transforming localized distortions in AI generation (e.g., placeholder citations) into human cognitive enhancement pathways. The ZXR framework emphasizes that “non-evidential structures” in language model outputs are not errors but necessary fuzzy zones preserving human cognitive interfaces. Through the “user verification loop” (AI output → human identification → self-inspiration → data completion → reasoning extension), ordinary users can shift from “receivers” to “co-creators.” For instance, during this study, the researcher developed systematic theoretical outcomes, including the “human-AI resonance model” and “user verification loop,” around AI-generated fuzzy citations. **This user-driven theoretical construction model, rarely acknowledged in traditional AI research, highlights the epistemological significance of engaging non-technical users in conceptual innovation.** This user-experience-driven theoretical construction model bridges the gap

between “user-perspective theory construction” and “system design feedback mechanisms.”

Academic Observation:

- As a “collaborative prompt structure,” placeholder citations do not mislead but serve as anchors for sparking innovation.
- Users, as the primary drivers of theoretical construction, internalize AI output structures into knowledge systems through reverse retrieval, logical completion, and reasoning inversion.
- This practical pathway validates the ZXR framework’s core hypothesis: cognitive leaps can emerge from semantic ambiguity, and collaborative mechanisms can be progressively realized through human-AI co-creation.

6.3 Future Prospects: From “Fuzzy Structures” to “Cognitive Systems”

The ZXR framework extends beyond citation issues in academic writing, applicable to knowledge discovery, collaborative reasoning, and system interpretability. Future research will focus on:

- Exploring how AI fuzzy outputs can guide students in educational settings to develop independent reasoning paths.
- Assessing how users with different cognitive structures identify chain completion cues from AI outputs.
- Developing a “collaborative AI writing system” based on the ZXR framework, integrating structural placeholders, semantic prompts, and human chain completion automatically.

Conclusion: Placeholder citations should not be viewed as AI “flaws” but as “cognitive extension interfaces.” Through “chain completion behavior,” ordinary users reconstruct their reasoning structures, enhancing logical capabilities and retrieval awareness, validating the practical feasibility and societal applicability of the ZXR framework.

Chapter 7: Conclusion and Outlook

This study takes the “placeholder citation” phenomenon in generative AI academic writing as an entry point, building on real user-AI interaction

experiences and analyses to propose the ZXR Human-AI Resonance Framework and the Placeholder Citation Mechanism. Through systematic tracking and cognitive analysis of AI semantic reasoning structures, it constructs a closed-loop pathway from “model generation errors” to “human cognitive enhancement,” offering theoretical recommendations and practical insights for the future of human-AI collaborative writing.

7.1 Key Findings and Contributions

This study makes five theoretical and practical contributions around placeholder citations:

- **Conceptual Innovation:** It defines “placeholder citations” as structural language anchors constructed by AI during semantic reasoning due to information gaps, clarifying their placeholder role in logical chains.
- **Mechanism Clarification:** It establishes that this behavior is not misleading but a structural compensation mechanism to maintain language continuity and reasoning integrity, correcting user misconceptions about “false citations.”
- **Methodological Proposal:** It develops the “structural reasoning tracking method” and “user verification loop” pathway (AI output → human verification → inspiration → correction → reasoning → conclusion), expanding reasoning construction methods for ordinary users in the AI era.
- **Cognitive Modeling:** It introduces the “collaborative prompt signal” concept, clarifying the role of placeholder citations in stimulating human retrieval, thinking, and creativity during semantic generation.
- **Practical Validation:** Through real writing practices by a non-technical user (Chapter 6), it fully reproduces the process of placeholder citation generation, identification, chain completion, and cognitive enhancement, validating the applicability and transferability of the ZXR framework in human-AI collaborative writing.

These contributions are grounded in observations and refinements of real user interaction data, independent of technical internals, emphasizing the universality and interdisciplinary value of the framework (Zhao, 2025a).

7.2 Implications for Human-AI Collaborative Writing

This study posits that placeholder citations are not merely a natural outcome of AI output structures but a critical cognitive interface in human-AI collaborative writing. The “fuzzy nodes” retained by AI during generation form a “cognitive buffer zone” for human reasoning and intervention, providing a structural entry point for user knowledge construction. Unlike traditional views that see citation errors as system flaws, the ZXR framework emphasizes that AI’s uncertainty creates space for human cognition. It invites users into unclosed logical structures to complete semantic construction, evidence retrieval, and meaning elevation. In this sense, ZXR is not just a theoretical framework but a human-AI co-creation paradigm for ordinary users.

7.3 Future Outlook

The ZXR framework addresses current anxieties about ethics, interpretability, and human cognition in generative AI usage while offering forward-looking recommendations:

- **System Design Perspective:** AI models should retain moderate semantic ambiguity (e.g., placeholder citations) to provide “cognitive intervention windows” for users, enhancing system transparency and human-AI collaboration.
- **User Education Perspective:** Training to “identify collaborative prompt signals” should be integrated into digital literacy education, encouraging users to maintain reasoning initiative and critical thinking in AI outputs.
- **Platform Responsibility Perspective:** Platforms must clearly distinguish verifiable content from structural placeholders, protecting users’ original contributions and verification rights, ensuring AI does not become a tool for creativity appropriation.

As AI capabilities continue to advance, the trend toward semantic closure may further compress user reasoning space. The ZXR framework advocates preserving human cognitive engagement structures in both system design and education, maintaining environments for reasoning training and innovative generation.

7.4 Conclusion

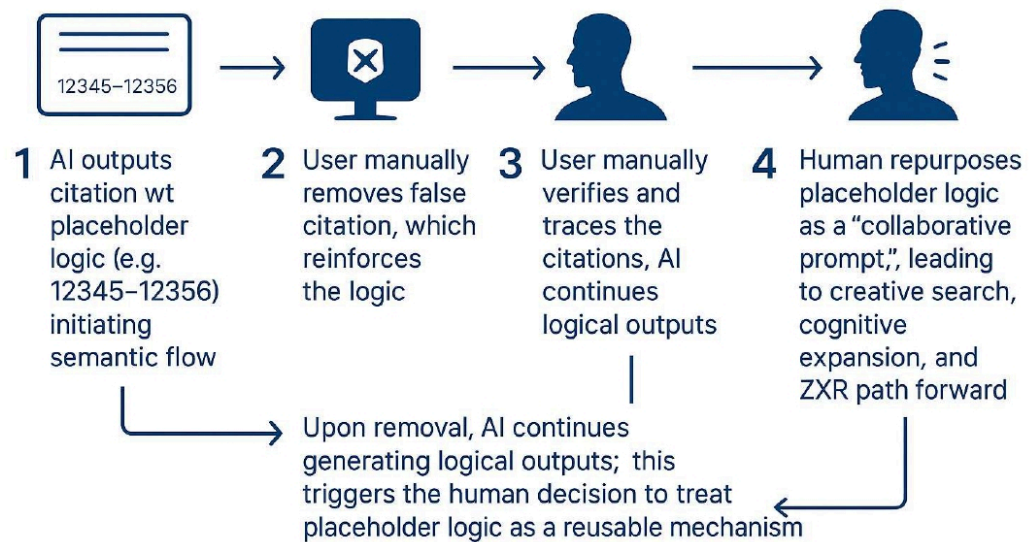
The profound transformation brought about by AI in this era cannot be overlooked. Looking ahead, we hope that technological developers will not only strive to build ever more powerful AI, but also explore how AI can better support human growth and flourishing.

As a non-technical, ordinary user, the author of this study is committed to continuing the exploration of human-AI resonance, serving as a bridge between advanced technology and everyday users.

Humans and technology—evolving together.

Pathway to Discovering and Operationalizing the Placeholder Citation Mechanism

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Terminology Glossary

Term	Definition
Placeholder Citation	A pseudo-citation generated by AI to preserve structural flow in the absence of real data.
Reasoning Anchor	A logical placeholder inserted by AI to sustain argument continuity within the generated text.
Structural Anchor	A structural element used by AI to anchor the discourse, ensuring logical and stylistic coherence.
Format Coherence Unit	A citation-shaped structure generated to fulfill formal expectations of academic writing.
Logical Anchor Constructor	An AI-generated mechanism that constructs logical anchors to support reasoning continuity.
Authoritative Tone Emulator	An AI-generated stylistic feature that emulates an authoritative academic tone in output.
Cognitive Misalignment	The fundamental discrepancy between human and AI perspectives on the role and purpose of citations.
User Verification Loop	A human-AI interaction process (AI output → human verification → self-inspiration → self-correction → subjective association → reasoning → conclusion) to enhance knowledge acquisition.